

Logistics

Online Learning

FTRL

MDPs

Introduction to RLRS: Online learning, FTRL, and Logistics

Carrie Chen and the CUAJ team

Logistics

This is a seminar that focuses on Reinforcement Learning. We care about: algorithm design, understanding algorithm performance (convergence) and the systems that make them scale in practice

1

Weekly meetings will be at 3:00pm EST Saturdays moving forward

Feedback form shared after today's seminar, please fill it out so we can make the next one better!

2

Paper voting system starting next week

3

Feel free to discuss in #weekly-papers when you have questions on the materials

That is the entire goal of this seminar!

4

RL Theory: AJKS is an official resource for this seminar

Online Learning

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MDPs

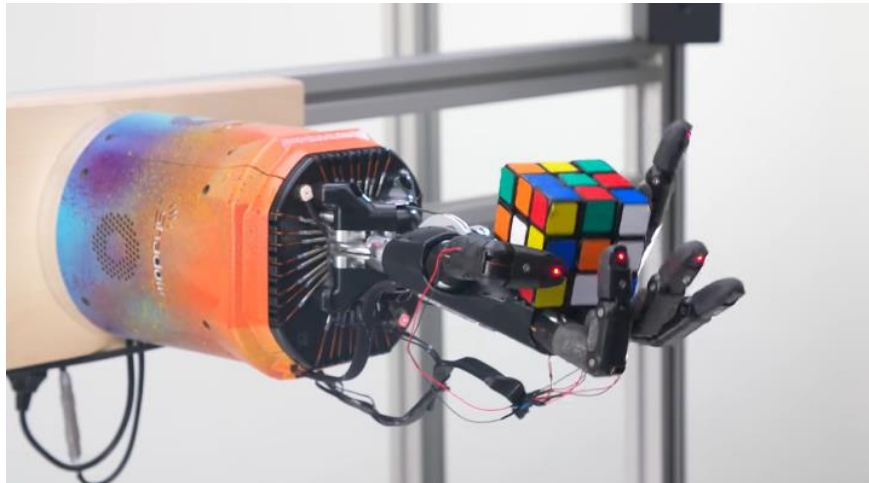
Big success in Reinforcement Learning



[AlphaZero, Silver et. AL, 17]



[OpenAI Five, 18]



[OpenAI Dactyl, 19]

Direct Preference Optimization: Your Language Model is Secretly a Reward Model

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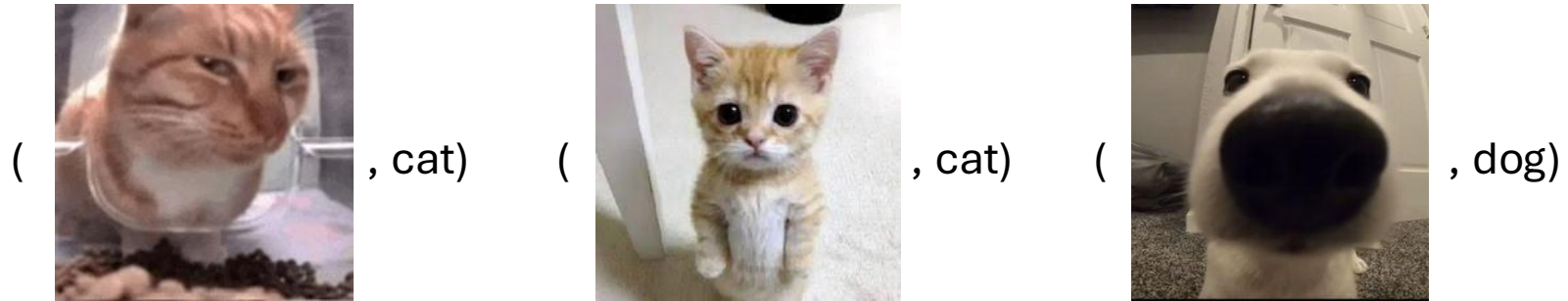
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Abstract

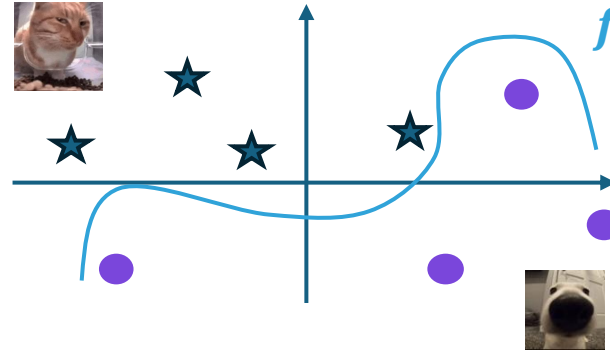
[Rafailov, 24]

Recap Supervised Learning

Given I.I.D. examples at training:



Fit a function on this stationary distribution of data:

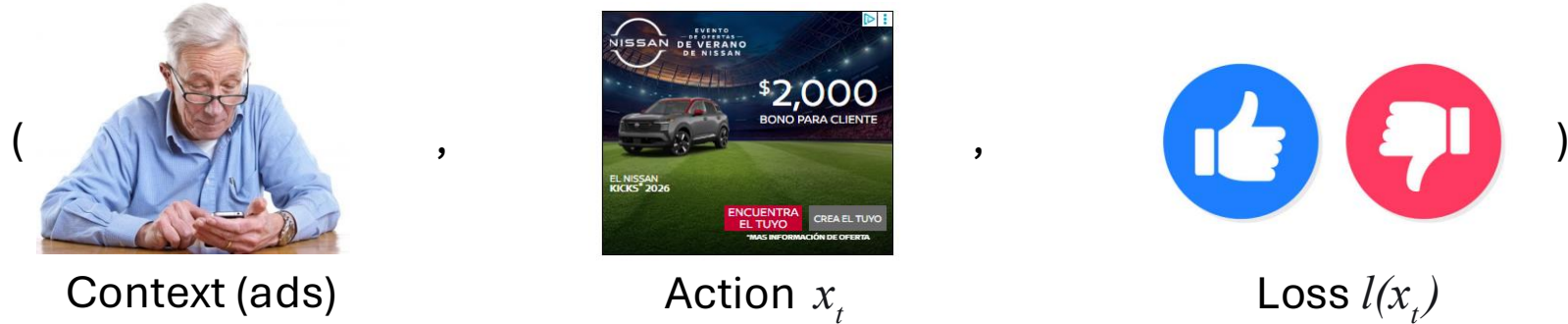


Deploy f into the real world, receive instant feedback

Passive: The outputs of f does not affect the incoming distribution of data it sees

Online (sequential) learning

Given a stream of non I.I.D. examples at time step t :



Instead of fitting once on a stationary distribution, actions depends on incoming sample + cumulative history:

$$x_{t+1} = \operatorname{argmin}_x \sum_{s=1}^t \ell_s(x)$$

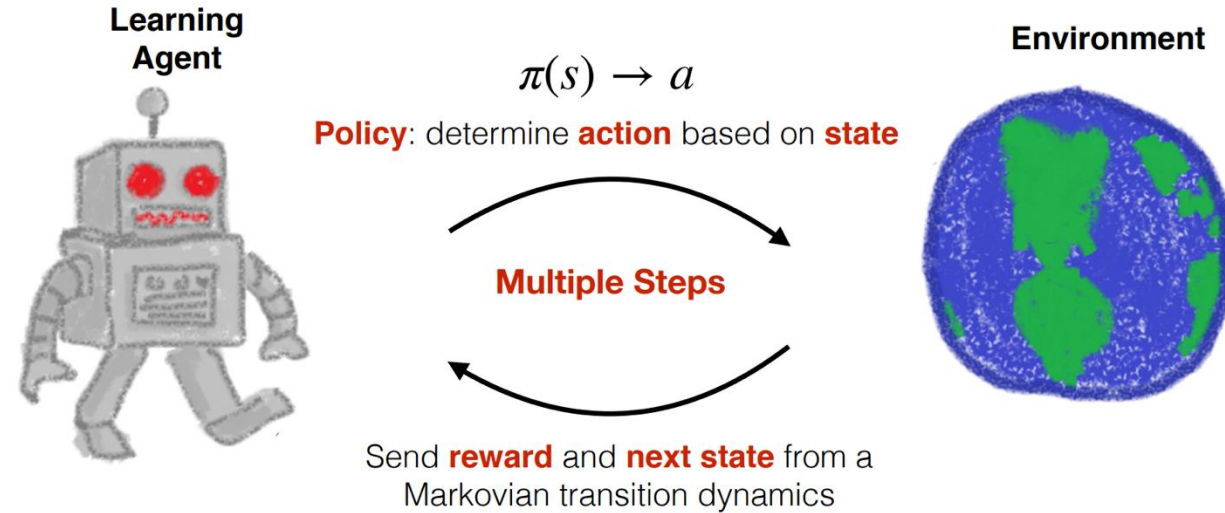
Performance is measured using $\text{Regret}_T = \underbrace{\sum_{t=1}^T \ell_t(x_t)}_{\text{what you actually paid}} - \underbrace{\min_{x^*} \sum_{t=1}^T \ell_t(x^*)}_{\text{best fixed point in hindsight}}$

Deploy into the real world, receive semi delayed feedback

Active: your actions may affect the incoming distribution of data seen next

Reinforcement Learning

Given a trajectory of non I.I.D. examples at time step t :



$$r(s, a), s' \sim P(\cdot | s, a)$$

Maximize for the discounted, cumulative reward: $\pi^* = \arg \max_{\pi} \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^T \gamma^t r(s_t, a_t) \right]$

Deploy into the real world, very delayed feedback

Active: your actions will affect the incoming distribution of data seen next

Summary so far

1. In RL, we often start from zero data/limited priors
2. In online and RL, our decisions have consequences: Future data is determined by our past decisions/predictions
3. To solve the task, we often need to make a long sequence of decisions

FTRL Terminology

Regret

How much worse we did than the **best fixed point chosen in hindsight**

$$\text{Regret}_T = \sum_{t=1 \dots T} f_t(x_t) - \min_x \sum_{t=1 \dots T} f_t(x)$$

“Low regret” = grows like $\sqrt{T}$ (*sublinear* $\sim \sqrt{T}$), so the **average** loss incurred per round $\rightarrow 0$ as T grows

Subgradient $g \in \partial f(x)$

A “slope” for functions with kinks, like $|x|$

$$f(y) \geq f(x) + g \cdot (y - x) \text{ for all } y$$

A line through $(x, f(x))$ that stays below the function. At the kink of $|x|$ every slope in $[-1, 1]$ qualifies. In the paper, $g_t = \nabla \ell_t(x_t)$ is the loss gradient

Q matrices

Generalized learning rates in matrix form

$$\frac{1}{2} \|Q_{1:t}^{\frac{1}{2}}(x - x_t)\|_2^2 = \frac{1}{2} (x - x_t)^\top Q_{1:t} (x - x_t)$$

$Q_{1:t}$ is **diagonal and adaptive**: coordinate i gets $(1/\gamma)\sqrt{(\sum_s g_{s,i}^2)}$ $\rightarrow$ frequently-updated features learn slower

Strong convexity

A class of policies that are easy to bound

$$f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \frac{\mu}{2} \|y - x\|^2$$

for all (x, y) in the domain, the function must lie above its tangent plane by at least a quadratic margin

Follow the Regularized Leader

The leader = the action/point that would have max-minimized regret so far

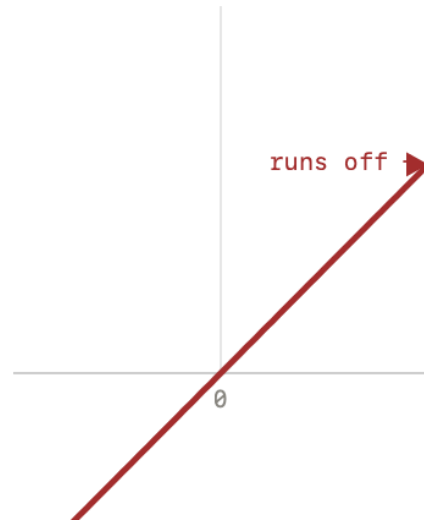
Follow-the-leader (naïve solve)

$$x_{t+1} = \arg \min_x g_{1:t}^\top x$$

Unbounded optimization problem,
argmin solver runs off into neg infinity
From $-g$ direction

FTL: plane

$\langle g, x \rangle \rightarrow$ no bottom



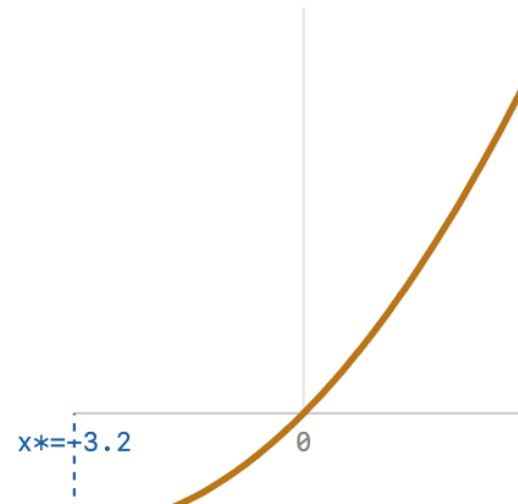
Follow-the-regularized-leader (quadratic reg)

$$x_{t+1} = \arg \min_x g_{1:t}^\top x + (\sigma_{1:t}/2) \|x\|_2^2$$

This is a **bowl** (the quadratic term) with a **tilt** (the linear term) --> a unique minimum exists! Set the gradient = 0: $x_{t+1} = -g_{1:t} / \sigma_{1:t} = -\eta_t g_{1:t}$

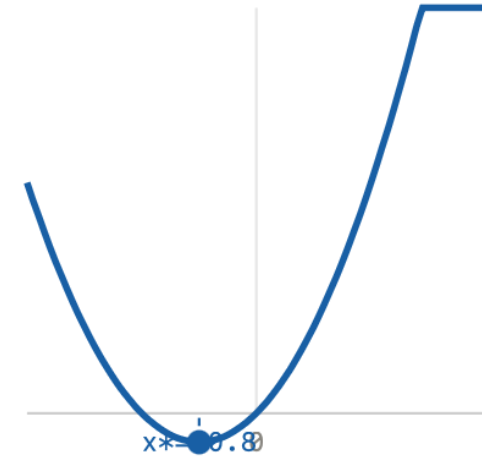
FTRL: weak σ

small $\sigma \rightarrow$ big shift
(only convex)



FTRL: strong σ

large $\sigma \rightarrow x^*$ near 0
(strongly convex)



Follow the Regularized Leader

An interactive, worked example: FTL vs FTRL on an adversarial opponent!

AN ONLINE LEARNING GAME / YOU ARE THE ADVERSARY

Beat the learner. *One of them is easy.*

Two algorithms split their bets between expert A and expert B. **FTL** goes all in on whichever expert has the lower total loss so far, and its choice is public. **FTRL** plays softmax of the same totals, so it hedges. Each round **you** choose where the loss lands. Your goal: run up their **regret**, meaning their loss minus the best single expert's loss.

round 0 FTL has committed to A (tie-break), and everyone can see it.

A

total loss $L_A = 0$

FTL, all in

50% of FTRL

⚡ Drop loss on A

B

total loss $L_B = 0$

50% of FTRL

⚡ Drop loss on B

FTRL's allocation 50% A / 50% B

Best expert's loss
0.0

FTL regret
0.0

FTRL regret
0.0

Your hit rate on FTL
n/a

The learners have committed. FTL's position is public information. Do your worst.

Demo and code here in the github: [rl-reading-group/demos/week_1_ftl/](https://github.com/rl-reading-group/demos/week_1_ftl/)*

Four Baseline Algorithms

The tradeoff: centering at 0 keeps updates close to what you've already committed to, while computing Ψ exactly captures the full regularization cost

Center convexity at current point $\mathbf{x}_s$

Center convexity at the origin 0

Approximate Ψ
using past sub-
gradients (ϕ_s)

FOBOS

= composite mirror descent

$$\mathbf{g}_{1:t} \cdot \mathbf{x} + \phi_{1:t-1} \cdot \mathbf{x} + \Psi(\mathbf{x}) + \frac{1}{2} \sum_s \|\mathbf{Q}_s^{\frac{1}{2}}(\mathbf{x} - \mathbf{0})\|_2^2$$

AOGD

adaptive online gradient descent

$$\mathbf{g}_{1:t} \cdot \mathbf{x} + \phi_{1:t-1} \cdot \mathbf{x} + \Psi(\mathbf{x}) + \frac{1}{2} \sum_s \|\mathbf{Q}_s^{\frac{1}{2}}(\mathbf{x} - \mathbf{x}_s)\|_2^2$$

Compute Ψ exactly,
cumulatively: $t \Psi(\mathbf{x})$

FTRL-Proximal

(this paper)

$$\mathbf{g}_{1:t} \cdot \mathbf{x} + t\Psi(\mathbf{x}) + \frac{1}{2} \sum_s \|\mathbf{Q}_s^{\frac{1}{2}}(\mathbf{x} - \mathbf{x}_s)\|_2^2$$

RDA

regularized dual averaging

$$\mathbf{g}_{1:t} \cdot \mathbf{x} + t\Psi(\mathbf{x}) + \frac{1}{2} \sum_s \|\mathbf{Q}_s^{\frac{1}{2}}(\mathbf{x} - \mathbf{0})\|_2^2$$

Every algorithm solves for $\mathbf{x}_{t+1} = \arg \min_{\mathbf{x}}$ using three-part objective: $\mathbf{g}_{1:t}$ (past gradients) + Ψ + strong convexity

Equivalence I: Proximal MD is proximal FTRL

Theorem 2. The local-looking step and the global optimization play **the same points**

Mirror descent (local view)

$$x_{t+1} = \arg \min_x g_t \cdot x + \Psi(x) + B_{1:t}(x, x_t)$$

“Fit the **latest** gradient, but don’t move far from where I am.”

=

FTRL-Proximal (global view)

$$x_{t+1} = \arg \min_x (g_{1:t} + \phi_{1:t-1}) \cdot x + \Psi(x) + \tilde{R}_{1:t}(x)$$

“Fit **all** past gradients at once, stabilized by quadratics centered at every past point.”

Why this matters: one step gradients may seem myopic but it actually has all the information we need

- 1 **Expand the Bregman term** $g_t^\top x + \Psi(x) + \tilde{R}_{1:t}(x) - \nabla \tilde{R}_{1:t}(\hat{x}_t)^\top x + \text{const}$
- 2 **Newest center kills its own gradient** $\nabla \tilde{R}_t(\hat{x}_t) = 0 \Rightarrow \nabla \tilde{R}_{1:t}(\hat{x}_t) = \nabla \tilde{R}_{1:t-1}(\hat{x}_t)$
- 3 **Last round’s optimality = the history** $-\nabla \tilde{R}_{1:t-1}(x_t) = g_{1:t-1} + \phi_{1:t-1}$ (induction: $x_t = \hat{x}_t$)
- 4 **Substitute FTRL appears** $(g_{1:t} + \phi_{1:t-1})^\top x + \tilde{R}_{1:t}(x) + \Psi(x) \quad \blacksquare$

Policy Optimization in FTRL Form

Policy iteration produces an advantage function $A_k^\pi(s, a)$. The policy update **is an online learning step**:

$$\pi_{k+1} = \arg \max_{\pi} \langle \pi, A_k^\pi \rangle - (1/\eta) D_{\text{KL}}(\pi \parallel \pi_k)$$

Local objective --> $g_{1:t} \cdot x$

Advantage estimates play the role of the accumulated gradients, the current best linear guess at improvement

KL penalty --> proximal R

Regularizes the update with KL distance from the previous time step, replacing the quadratic

Instances

TRPO, PPO, NPG, etc... all cautious regularized argmax steps in this format

Key takeaways

1. Regularization is SO important! FTRL's strong-convexity term exists precisely to keep each update close to what's already been learned
2. FTRL's guarantee is about cumulative regret over the whole sequence, not any one round. This is similar to the credit-assignment problem in RL: attributing reward at the end (retrospectively) across every decision that led there